

[1] Before entering a reactor, a gas stream of 50 mol% methane (CH₄) and 50 mol% nitrogen (N₂) must be heated from 20°C to 300°C. The gas stream enters a well-insulated heater at a flow rate of 2000 L(STP)/min. Assume ideal gas behavior. Data: $c_{p,CH_4} = 18.4 + 56.0 t$ and $c_{p,N_2} = 28.1 + 3.2 t$ where the heat capacity has units of J/mol K and t is temperature (K)/1000. ~~Assume constant pressure of 1 atm.~~

(a) Draw a schematic of the process. Calculate the heating rate in kilowatts required for the process. Do not neglect the temperature dependence of the heat capacities. (25 pts.)

$\dot{V}_i = 2000 \text{ L(STP)/min}$
 $X_{CH_4} = 0.5$
 $X_{N_2} = 0.5$

heater

$\dot{V}_e = ?$
 $X_{CH_4} = 0.5$
 $X_{N_2} = 0.5$

$\dot{n}_i = \left(\frac{\dot{V}_i}{V_{STP}} \right) = \frac{2000 \text{ L(STP)/min}}{22.4 \text{ L(STP)/mole}} = 89.29 \text{ mol/min} \text{ (5)}$

ignore mech. energy and work

$\dot{Q} = \Delta \dot{H} = \dot{n} \Delta \hat{H} = \dot{n}_{CH_4} \int_{20^\circ C}^{300^\circ C} c_{p,CH_4} dT + \dot{n}_{N_2} \int_{20^\circ C}^{300^\circ C} c_{p,N_2} dT$
 $\dot{n}_{CH_4} = \dot{n}_{N_2} = \frac{\dot{n}}{2}$

$\dot{Q} = \dot{n} \int_{293K}^{373K} \left(\frac{c_{p,CH_4} + c_{p,N_2}}{2} \right) dT = \dot{n} \int_{293K/1000}^{373K/1000} \left(\frac{18.4 + 28.1}{2} + \frac{56.0 + 3.2}{2} t \right) d(t/1000) * 1000$

$= \dot{n} \int_{0.293}^{0.573} (23.15 + 29.6 t) dt * 1000$

$= 1000 \dot{n} \left[23.15(0.573 - 0.293) + \frac{29.6}{2} (0.573^2 - 0.293^2) \right] \frac{J}{mol K}$

$= 1000 (89.29 \frac{mol}{min}) [6.482 + 14.8(0.24248)] \frac{J}{mol K} \text{ (5)}$

$= 1000 (89.29) [10.07] \frac{J}{min}$

$= 8.99,213 \frac{J}{min} \frac{min}{60s} \frac{KJ}{1000J} \text{ (5)}$

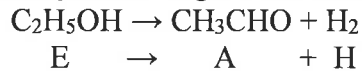
$= 15.0 \frac{KJ}{s}$

$\dot{Q} = \underline{15.0} \text{ kW}$ (Be sure to give the correct sign on your answer.)

(b) The actual required heat is 12% larger in magnitude than you calculated. Assuming all the values and data given are accurate and the gas behaves ideally, give two plausible reasons that explain this difference. (5 pts.)

$\Delta \dot{E}_K > 0$ / not at steady state /

[2] Ethanol (C_2H_5OH) can be dehydrated to give acetaldehyde (CH_3CHO) and hydrogen (H_2).

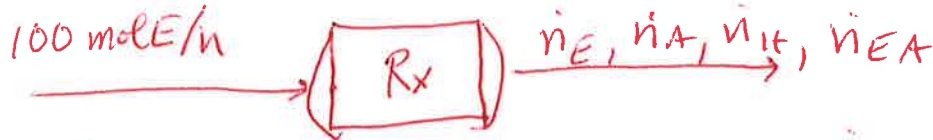


An undesired reaction also occurs that produces ethyl acetate ($CH_3COOC_2H_5$) and hydrogen (H_2).



Ethanol enters a reactor at a flow rate of 100 mol/h. The fractional selectivity (as defined on page 299 of the textbook) of the acetaldehyde is 65%. The ethyl acetate is 9 mol% of the exiting stream from the reactor.

(a) Calculate the molar flow rates in mol/h of all the species in the gas stream leaving the reactor. (35 pts.)



(10) $\dot{n}_{E,out} = 100 - \dot{\xi}_1 - 2\dot{\xi}_2$

$$\dot{n}_{A,out} = \dot{\xi}_1$$

$$\dot{n}_{H,out} = \dot{\xi}_1 + 2\dot{\xi}_2$$

$$\dot{n}_{EA,out} = \dot{\xi}_2$$

$$\dot{n}_{tot,out} = 100 + \dot{\xi}_1 + \dot{\xi}_2$$

Selectivity $f_s = \frac{\dot{n}_{A,out}}{\dot{n}_{E,in} - \dot{n}_{E,out}} = \frac{\dot{\xi}_1}{\dot{\xi}_1 + 2\dot{\xi}_2}$

$x_{EA,out} = \frac{\dot{n}_{EA,out}}{\dot{n}_{tot,out}} = \frac{\dot{\xi}_2}{100 + \dot{\xi}_1 + \dot{\xi}_2}$ (10)

$$(0.09)(100 + \dot{\xi}_1 + \dot{\xi}_2) = \dot{\xi}_2$$

$$9 = -0.09\dot{\xi}_1 + 0.91\dot{\xi}_2$$

$$\dot{n}_{E,out} = 100 - 58.06 - 2(15.63) = 10.67 \frac{mol}{h}$$

$$\dot{n}_{A,out} = 58.06 \frac{mol}{h}$$

$$\dot{n}_{H,out} = 58.06 + 2(15.63) = 89.33 \frac{mol}{h}$$

$$\dot{n}_{EA,out} = 15.63 \frac{mol}{h}$$

(5)

$$(0.65)(\dot{\xi}_1 + 2\dot{\xi}_2) = \dot{\xi}_1$$

$$0 = 0.35\dot{\xi}_1 - 1.3\dot{\xi}_2$$

$$\dot{\xi}_1 = \frac{1.3}{0.35}\dot{\xi}_2$$

$$9 = -(0.09)\left(\frac{1.3}{0.35}\right)\dot{\xi}_2 + 0.91\dot{\xi}_2$$

$$\dot{\xi}_2 = \frac{9}{0.91 - (0.09)\left(\frac{1.3}{0.35}\right)} = 15.63 \frac{mol}{h}$$

$$\dot{\xi}_1 = \left(\frac{1.3}{0.35}\right)(15.63 \frac{mol}{h}) = 58.06 \frac{mol}{h}$$

(10)

Atomic Balance (alternate solution)

$$C: 2\dot{n}_{E,in} = 2\dot{n}_{E,out} + 2\dot{n}_{A,out} + 4\dot{n}_{EA,out} \parallel \frac{1}{2} \quad \text{oxygen balance same as carbon}$$

$$H: 6\dot{n}_{E,in} = 6\dot{n}_{E,out} + 4\dot{n}_{A,out} + 2\dot{n}_{H_2,out} + 8\dot{n}_{EA,out} \parallel \frac{1}{2}$$

$$65 = 0.65\dot{n}_{E,out} + \dot{n}_{A,out}$$

$$\Leftrightarrow 0.65 = \frac{\dot{n}_{A,out}}{100 - \dot{n}_{E,out}}$$

$$0 = \dot{n}_{E,out} + \dot{n}_{A,out} + \dot{n}_{H_2,out} - 10.111\dot{n}_{EA,out} \Leftrightarrow 0.9 = \frac{\dot{n}_{EA,out}}{\dot{n}_{E,out} + \dot{n}_{A,out} + \dot{n}_{H_2,out} + \dot{n}_{EA,out}}$$

$$\begin{pmatrix} \dot{n}_{E,out} \\ \dot{n}_{A,out} \\ \dot{n}_{H_2,out} \\ \dot{n}_{EA,out} \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 & 2 \\ 3 & 2 & 1 & 4 \\ 0.65 & 1 & 0 & 0 \\ 1 & 1 & 1 & -10.111 \end{pmatrix} \begin{pmatrix} 100 \\ 300 \\ 65 \\ 0 \end{pmatrix} = \begin{pmatrix} 10.67 \\ 58.06 \\ 89.33 \\ 15.63 \end{pmatrix} \text{ mol/h}$$

Same results as before!

$$\dot{n}_E = \underline{10.67} \text{ mol/h}, \dot{n}_A = \underline{58.06} \text{ mol/h}, \dot{n}_H = \underline{89.33} \text{ mol/h}, \dot{n}_{EA} = \underline{15.63} \text{ mol/h}$$

(b) Calculate the fractional conversion of ethanol to products and byproducts. (5 pts.)

$$f_{C,E} = \frac{\dot{n}_{E,in} - \dot{n}_{E,out}}{\dot{n}_{E,in}} = \frac{100 - 10.67}{100} = 0.8933$$

$$f_{C,E} = \underline{0.8933}$$

[3] A reactor is fed 2 mol/s of A, 1 mol/s of B and 3 mol/s of I an inert gas at a total pressure of 1 atm absolute and 25°C. The product C forms according to the chemical reaction: $A + B \rightarrow C$
 The reaction takes place in the gas phase. Assume ideal gas behavior. The equilibrium expression is given by $K = \frac{y_C}{y_A y_B} \left(\frac{P_0}{P(atm)} \right)$ where $P_0 = 1$ atm and $P(atm)$ is the pressure in atmospheres absolute and $K=1.7$ is the equilibrium constant.

(a) What is the extent of reaction in moles/s if the product stream exits at equilibrium? (25 pts.)

$$\begin{aligned} n_{A,out} &= 2 - \xi \\ n_{B,out} &= 1 - \xi \\ n_{C,out} &= \xi \\ n_{I,out} &= 3 \\ \hline n_{tot,out} &= 6 - \xi \end{aligned}$$

$$\begin{aligned} K=1.7 &= \frac{\xi(6-\xi)}{(2-\xi)(1-\xi)} \left(\frac{1 \text{ atm}}{1 \text{ atm}} \right) \\ 1.7(2-\xi)(1-\xi) &= \xi(6-\xi) \\ 0 &= 1.7(2-3\xi+\xi^2) - (6\xi-\xi^2) \\ 0 &= 3.4 - 11.1\xi + 2.7\xi^2 \\ \xi &= \frac{11.1 \pm \sqrt{(11.1)^2 - 4(3.4)(2.7)}}{2(2.7)} \\ &= \frac{+11.1 \pm 9.3}{2(-2.7)} = 0.33\bar{3} = \frac{1}{3} \end{aligned}$$

$$\xi = 0.33\bar{3} \text{ mol/s} \quad \left(\frac{1}{3} \right)$$

(b) Would increasing the flow rate of the inert while maintaining constant pressure and temperature in the reactor, increase, decrease, or have no effect on the equilibrium extent of reaction? To receive credit, you must clearly justify your answer. (5 pts.)

$$\frac{\xi(3 + \dot{n}_I - \xi)}{(2-\xi)(1-\xi)} = 1.7$$

$y_A, y_B, \text{ and } y_C \text{ all get diluted}$

$$K = \frac{y_C}{y_A y_B}$$